

Model Cards and Human Review

Project: Responsible Plant Classifier

1. Model Cards and Human Review

Teach the learning cycle

Primary target: Write a model card whose claims match evidence

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Proxy Features and Fairness

Without opening the old lesson, explain one key idea from Proxy Features and Fairness. Give an example and a mistake that the idea helps you avoid.

Look for: Identify possible proxy features

Reference example, not the only acceptable wording: Teacher explanation: X occurs 8/10 north and 2/10 south in this sample. This supports an association, not a deterministic identity rule or a causal explanation.

Improve Generalization

Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

Look for: Compare training and validation evidence for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

Overfitting

Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

Look for: Compare training and validation evidence for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve training MAE 0 does not compensate for validation MAE 8.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Draft five model-card headings for a classroom leaf classifier. Why must the card state that its labels are classroom categories rather than professional plant diagnosis?

Concept to explain

A model card defines intended use, evidence, limitations, risks, decision boundaries, monitoring, and human escalation.

Term · Meaning

Term: model card

Meaning: deployment-facing model documentation

Term: intended use

Meaning: approved purpose and setting

Term: do-not-use

Meaning: explicit prohibited application

Term: human review

Meaning: authorized person checks or decides

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on model cards and review. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer. Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Readiness answer

Data provenance, model version, full results, limitations and decision history.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank evidence table

Saved classroom project records for critique; no personal student dataset required

End goal for the pictorial case

By the end, I can explain model cards and human review using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Write a model card with a clear do-not-use case and review path.

MODEL IDEA A model card defines intended use, evidence, limitations, risks, decision boundaries, monitoring, and human escalation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Intended use: classroom exploration of visible leaf spotting. Evidence: test metrics, confusion matrix, subgroup slices, and data version.

Draw a fresh data → model → prediction → evaluation trace:

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

STEP 1 Intended use: classroom exploration of visible leaf spotting.

STEP 2 Evidence: test metrics, confusion matrix, subgroup slices, and data version.

STEP 3 Do not use for pesticide, medical, or agricultural diagnosis.

STEP 4 Uncertain, unfamiliar, or high-consequence cases route to a teacher or qualified adult.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

Concrete teaching case

A model card says healthy/spotted, 80% accuracy only.

Add responsible-use evidence.

Exact explanation to model

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and human review. Classroom classification is not professional diagnosis.

Explain the added worked example

A model card helps another person understand what was built and what evidence exists. It is a record, not a certificate of safety. Link statements to saved data and run records. Use unknown or not evaluated where evidence is absent instead of filling gaps with optimistic language.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines model cards and review.

Example card: classroom parcel classifier v 1; purpose: compare toy classification methods on fictional clear/blocked labels. Training 40, validation 20, test 12 with separate record groups. Frozen test TN 7 FP 1 FN 2 TP 2, blocked positive: accuracy 9/12, precision 2/3, recall 2/4. Always-clear baseline scores 8/12. No real parcel or student decision is supported.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Key terms

Term · Meaning

Term: Model card

Meaning: A concise structured account of a model and its limits

Term: Intended use

Meaning: The purpose and conditions the evidence supports

Term: Out-of-scope use

Meaning: A purpose for which the model is not supported

Term: Review status

Meaning: The current state of checks and unresolved issues

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Model cards and review at a glance

Part · Evidence or action · Interpretation

Part: Purpose and version

Evidence or action: Toy classifier v 1

Interpretation: Fictional data only

Part: Evidence

Evidence or action: 12 held-out cases

Interpretation: Full matrix and baseline

Part: Limits and review

Evidence or action: No real decisions

Interpretation: List unresolved coverage

The model card should name feature availability, preprocessing fitted on training, model settings, reference-label rules and test date. Include subgroup coverage if evaluated; otherwise say it was not assessed. Describe relevant failure examples without exposing personal information.

Verify the worked accuracy, precision and recall. Explain the steps or evidence behind your response.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Model Cards and Human Review

Write a model card whose performance statement can be traced to a concrete evaluation.

The facts we will use

Toy $v = 3$ classifier scored 16/20 on five unseen plants with four views each, under indoor tablet lighting. Baseline scored 14/20.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Model Cards and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name evidence

Record $v = 3$, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.



2. Compare baseline

$v = 3$ accuracy 80%; baseline 70% on the same 20 cases.



3. State limits

Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

CHANGE ONE THING

The publicity sentence becomes "80% accurate for every plant".

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Name evidence

What happens and why: Record $v = 3$, 20 photo cases, 5 plant identities, group-separated split and indoor tablet conditions.

Step: 2. Compare baseline

What happens and why: $v = 3$ accuracy 80%; baseline 70% on the same 20 cases.

Step: 3. State limits

What happens and why: Do not claim all cameras, outdoor conditions or disease diagnosis were validated.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Do not claim all cameras, outdoor conditions or disease diagnosis were validated. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A model card communicates evidence; it is not certification of safety, fairness or suitability for consequential plant decisions.

Change one thing in the pictured case

Change: The publicity sentence becomes “80% accurate for every plant”.

Prediction: Reject that unbounded statement.

Reason: The measured rate belongs to twenty related indoor photos, not every plant or setting.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A model card says healthy/spotted, 80% accuracy only.

Write the question to explain: Add responsible-use evidence.

Reveal after predictions: Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and human review. Classroom classification is not professional diagnosis.

Keep for an independent attempt: A model card says 90% accuracy but gives no intended use, data conditions or human-review rule. What should be added before classroom use?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided response: $TN / 7$ $TP / 2$ give $9/12$ accuracy; $TP / (TP + FP) = 2/3$ precision; $TP / (TP + FN) = 2/4$ recall. Exit response: Record a new version, threshold, reason, affected evaluations and review status; retain prior results and distinguish new independent evidence from reused development data.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Use the prerequisite question: What should remain visible after an evaluation? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify model card in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: draft a purpose; check arithmetic. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between model card and intended use. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: A card says fair and reliable but provides only accuracy. Request three additions.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Write a model card whose claims match evidence; Separate intended use from unsupported use. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Completion task: Verify the worked accuracy, precision and recall.

Teacher check: Teacher explanation: $TN / 7$ $TP / 2$ give $9/12$ accuracy; $TP / 2 / (TP / 2 + FP / 1) = 2/3$ precision; $TP / 2 / (TP / 2 + FN / 2) = 2/4$ recall.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one

step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Model cards and review. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Intended: compare toy classifiers on the documented fictional task. Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Write a bounded replacement for the publicity sentence. Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

“ $v = 3$ matched 16 of 20 reference labels on five held-out plants photographed indoors by tablet; baseline matched 14 of 20 under the same conditions.” Changed-case check: Reject that unbounded statement. The measured rate belongs to twenty related indoor photos, not every plant or setting.

Changed-case reasoning: Reject that unbounded statement. The measured rate belongs to twenty related indoor photos, not every plant or setting.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected evidence before running.
2. Model Card Clinic: compare two cards, find unsupported claims and missing gates, then author a release-ready version.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

MISCONCEPTION TO REPAIR A disclaimer at the bottom of the screen is a complete safety plan.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Intended: compare toy classifiers on the documented fictional task. Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

A stronger claim makes a stronger project: Record a new version, threshold, reason, affected evaluations and review status; retain prior results and distinguish new independent evidence from reused development data.

The new case can be copied from the example: The independent task changes the evidence. Require a fresh explanation: Request evaluated counts, split history, reference labels, baseline, errors and supported use. Replace the universal claim with: performance scope is not established from the supplied evidence; the reported 90 percent cannot yet be audited.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Intended users/conditions, labels/data, evaluation sets/metrics, class/subgroup results, versions, limitations, excluded uses and human review. Classroom classification is not professional diagnosis.

Independent question

A model card says 90% accuracy but gives no intended use, data conditions or human-review rule. What should be added before classroom use?

Expected independent answer

Add intended and excluded uses, dataset and evaluation conditions, subgroup results and limitations, plus when uncertain or consequential cases go to a human.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “One aggregate score is sufficient documentation for responsible use.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Include intended use/users, data and label definitions, model/version, evaluation and subgroup results, limitations/excluded uses, and human-review process. The card should connect claims to recorded tests and avoid presenting classroom labels as professional advice.

Explanations for the additional workbook examples

W1 Draft a purpose

Write an intended-use and non-use statement for the worked card.

Teacher explanation: Intended: compare toy classifiers on the documented fictional task. Not supported: real parcel handling, student assessment or decisions outside the evaluated setup.

W2 Check arithmetic

Verify the worked accuracy, precision and recall.

Teacher explanation: TN 7 TP 2 give 9/12 accuracy; $TP\ 2 / (TP\ 2 + FP\ 1) = 2/3$ precision; $TP\ 2 / (TP\ 2 + FN\ 2) = 2/4$ recall.

W3 Find missing evidence

A card says fair and reliable but provides only accuracy. Request three additions.

Teacher explanation: Request full error counts and denominators, relevant subgroup coverage/results, data provenance and limits; remove unsupported fairness/reliability claims.

W4 Apply independently

Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

Teacher explanation: Request evaluated counts, split history, reference labels, baseline, errors and supported use. Replace the universal claim with: performance scope is not established from the supplied evidence; the reported 90 percent cannot yet be audited.

W5 Maintain the record

What should change in a card after a threshold revision?

Teacher explanation: Record a new version, threshold, reason, affected evaluations and review status; retain prior results and distinguish new independent evidence from reused development data.

Assess independent transfer

Selected case: Review this card: model v 2, 90 percent accurate, works for every school. No dataset size, split or baseline is given. Write a correction request with a bounded replacement claim.

Selected case answer

Teacher explanation: Request evaluated counts, split history, reference labels, baseline, errors and supported use. Replace the universal claim with: performance scope is not established from the supplied evidence; the reported 90 percent cannot yet be audited.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now

and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Responsible Plant Classifier

Complete the Plant Classifier model card and require teacher sign-off before any classroom demonstration.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 34: Capstone Dataset and Baseline. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide a blank evidence table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use model card and intended use in full sentences. Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Request evaluated counts, split history, reference labels, baseline, errors and supported use. Replace the universal claim with: performance scope is not established from the supplied evidence; the reported 90 percent cannot yet be audited. Fresh retry: a card lists test accuracy 18/20 but omits positive-class recall and intended use. Ask for full matrix, class definition and bounded use; do not infer recall from accuracy. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Capstone data and requirements. Carry forward the saved evidence and the distinction between development choices and justified claims.

Plan the delayed return

Suggested return points in this level: ai-4/34, ai-4/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.